Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Review - Instantaneous Present

Benjamin D. Summers <bsummers@adagiollc.com> Fri, Sep 18, 2026 at 2:22 AM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>
Cc: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Sara,

Thank you for going through the responses. The most recent revision of the manuscript is now at https://dx.doi.org/10.2139/ssrn.6922380; it carries the changes that address your review, Ezequiel's, and your two remaining points:

Point 4. Accepted. The sentence you quote was in my email and not in the paper. Sections 2 and 6 say "generically" and cite Burke-Rosenblatt and Kemeny-Snell for that statement only. Two sentences in the paper did state the generic failure as universal, one in Section 2 and one in Section 6; both now say "generically". The two-state calculation is not in the paper. In its place, Lemma 1 in Section 6 proves the linear scaling for a class: for an irreducible finite-state chain in its stationary law, observed through a window of width δ, every row of the composition defect of the window average has total variation at most $C\, q_* \delta$, with q_* the maximal jump rate, $C = 2 + 8/c$ explicit in the generator and the lag, and $\delta \le \min\{(t - t_1)/2, 1/(2q_*)\}$. The abstract and the test in Section 6 had stated that rate for any process that is Markov on the observed state space; they now carry the lemma's hypotheses. The two-state chain was the check I ran on the lemma, and the paper does not report it.

Point 12. Accepted. Section 3 says the operational result does not establish that the mathematical state at the point is incomplete and that no such claim is made; Section 4 says the argument is evidential, not deductive; Section 7 now presents the reversal of the burden of proof as the conclusion argued, not as a consequence of the three physical arguments alone. Your formulation is adopted as its operational form. Section 7 now says that memorylessness is not to be presumed without a test of its stability under refinement of the temporal resolution and under changes of the state-space specification, and that Section 6 supplies the test. The two halves of Lemma 1 are those two changes: refinement of the resolution at fixed state space, where the defect vanishes linearly, and a change of the state space by a non-lumpable map at any resolution, where the defect converges to the lumped family's and does not vanish.

Let me know your thoughts.

Ben

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On Wed, Sep 16, 2026 at 8:18 AM ISMAIL-SUTTON, SARA R. <sara.r.ismail-sutton@durham.ac.uk> wrote:

> Hi Ben,
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> Thank you for the detailed response. I have gone through your explanations, as promised.
>
> The revisions you describe in points 1–3 and 5–11 appear to be directed appropriately at the concerns raised in my report. As I have not reviewed the revised manuscript itself, I cannot confirm how fully those points have been resolved in the text, but I do not have any further comments on your descriptions of those changes at this stage.
>
> There are two remaining conceptual points arising from your response.